

«Approved»
Acting Dean of the School of Intelligent Systems

L.S. Aldasheva
«01» 09 2026 y.

Syllabus
Academic Year 2026 - 2027



1. General Information							
Course Code	DLD5						
Course title	Digital Logic Design						
Degree Cycle (Level)/ Major / Relation to curriculum	Bachelor's Degree 6B06301 - Cybersecurity						
Year, term	2 nd year, 4 th trimester						
Language of Instruction:	English						
Lecturer(s)	Alisher Amirov, MSc in Information Technologies, Senior Lecturer at the School of Intelligent Systems alisher.amirov@astanait.edu.kz Office: 1.1.320 Alisher Batkuldin, MSc in Electrical and Computer Engineering, Senior Lecturer at the School of Intelligent Systems a.batkuldin@astanait.edu.kz Office: C1.1.221 Aigerim Kalikova, MSc in Engineering, Senior Lecturer at the School of Intelligent Systems aigerim.kalikova@astanait.edu.kz Office: C1.1.111 Didar Yarulin, MSc in Physics, PhD Candidate in Automation and Control, Senior Lecturer at the School of Intelligent Systems D.yarulin@astanait.edu.kz Office: 1.1.111 Olzhas Konakbayev, MSc in Automation and control, Senior Lecturer at the School of Intelligent Systems o.konakbayev@astanait.edu.kz Office: 1.1.111						
Number of credits	5						
Workload of course components and credits per trimester							
	ECTS credits	Contact hours			IASS	SIS	Total hours
		Lectures	Practice sessions	Labor atory work			
		5	30	20			
Prerequisites	Mathematics for Computer Science, Introduction to Computer Science, Programming Fundamentals						
Postrequisites	Computer Architecture for Cybersecurity track Introduction to Microcontroller programming, Embedded systems design, Sensors for IoT						
2. Goals, Objectives and Learning Outcomes of the Course							
1. Course Overview/Description	This course introduces sophomore year students to the fundamental principles of digital design, focusing on the creation and analysis of digital circuits and systems. Students will develop a strong foundation in digital logic and gain hands-on experience using industry-standard digital design tools. The course also explores the integration of digital design concepts with intelligent systems and cybersecurity.						
2. Course Goal(s)	Learning	Course goal is to provide a solid background in digital logic design via problem-solving and laboratory practices					

This individual opportunity can be considered by teacher in case if student can not to present approved medical document from medical policlinics on any reasons.

In any other cases **student must attend to the course not less 70% of all classes.**

Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the **QUALITY** of their contributions and insights. Quality comments possess one or more of the following properties:

- Offers a different and unique, but relevant, perspective;
- Contributes to moving the discussion and analysis forward;
- Builds on other comments.

Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student can be dispensed from the classes.

Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the teacher.

Attestation I and II: Students with score less than 25% for Attestation period I or Attestation period II (RK1/RK2) are automatically failed and should take the course again.

Home work / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures.

Late assignments: Late assignments: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale.

In the case of some extraordinary event, students should notify the teacher and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances by teacher.

Final exam / Final Project: The final exam for the course "Digital Logic Design" includes multiple-choice test for one hour which covers theoretical and practical parts of the course. At the completion of the exam, all works must be submitted in the Learning Management System (lms.astanait.edu.kz). No late submissions are allowed in the exam. *Format of exam (midterm, end term and final) may vary.*

Laptops and mobile devices can only be used for classroom purposes when directed by the teacher. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the teacher.

Online lessons can be used in case if there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (lms.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround.

Cheating and plagiarism are defined in the Academic conduct policies of the university and include:

1. Submitting work that is not your own papers, assignments, or exams;
2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation;

	3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (lms.astanait.edu.kz). Contacting the Instructor (Teacher): The easiest and most reliable way to get in touch with the teacher is by email. Students must feel free to send email if you have a question related to the course. The teachers will respond as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the teacher by their office during office hours to discuss the class using both offline and online ways.
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3.1 Abbreviations

#	Abbreviation	Meaning
1	TSIS	Teacher-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.2 Course Plan

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	Ch1 – Digital Electronics Intro Ch2 – Numbers we use in Digital Electronics Conversions, Complements	3		2		
2	Ch3 – Logic Gates	3		2		
3	Ch4 – Combining Logic Gates Maxterms, minterms, Boolean algebra, Karnaugh maps, Karnaugh maps with three variables Karnaugh maps with four variables Karnaugh maps with five variables De Morgan's theorems	3		2		
4	Ch6 – Encoding, Decoding	3		2		
5	Ch7 – Flip-flops (Midterm)	3		2		
6	Ch8 - Counters	3		2		
7	Ch9 – Shift Registers	3		2		
8	Ch10 – Arithmetic circuits Binary Addition, Half Adders, Full Adders, 3-bit Adders, Binary Subtraction, Binary Multiplication, 2s Complement notation, addition, subtraction, 2s Complement Adders/Subtractors	3		2		
9	Ch11 – Memories RAM, ROM	3				
10	Ch13 – Computer Systems (Endterm)	3		2		

	Total hours: 50	30		20		
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4. Student performance evaluation system for the course.

Period	Assignments	Number of points	Total
1 st attestation	Assignment 1 Assignment 2 Assignment 3 Midterm	20 20 20 40	100
2 nd attestation	Assignment 4 Assignment 5 Assignment 6 Endterm	20 20 20 40	100
Final exam	Multiple Choice Questions or Final Project		100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final		100

**Each assignment is combination of tasks including labs, quizzes practical assessments and other forms of submission depending on the topic the week.*

***Students who miss lessons due to health issues must provide an official statement from the hospital or clinic as well as from the dean's office. They must also complete all missed assignments and tasks by the date indicated on the statement.*

Course Assessment Rubric for Student Performance Evaluation

Learning outcomes	Indicators	Excellent (90-100)	Good (70-89)	Satisfactory (50-69)	Fail 0-49
Number systems and digital logic fundamentals: number representation, digital electronics, logic levels, logic gates, Boolean algebra	1. Can clearly explain number systems and basic digital logic concepts. 2. Uses correct terminology when describing logic gates and Boolean operations. 3. Can construct and interpret truth tables.	Demonstrates comprehensive understanding of number systems, digital electronics, logic gates, and Boolean logic; accurately analyzes and simplifies logical expressions; independently solves complex problems.	Shows solid understanding of digital logic concepts; correctly analyzes logic gates and Boolean expressions; solves problems with minor errors and minimal guidance.	Demonstrates basic understanding of number systems and logic gates; can construct simple truth tables and perform basic Boolean operations, but makes frequent errors.	Lacks fundamental understanding of number systems and digital logic; unable to construct or interpret basic truth tables or apply Boolean operations.
Combinational logic circuits: combining logic gates, IC specifications and interfacing, encoding and decoding, arithmetic circuits	1. Can analyze and design combinational logic circuits. 2. Can select and use appropriate digital ICs. 3. Can implement encoders, decoders, and arithmetic circuits.	Demonstrates comprehensive understanding of combinational logic; independently designs, implements, and tests reliable circuits; correctly selects ICs and analyzes datasheet specifications; solutions are optimized and well documented.	Shows solid understanding of combinational circuits; can design and implement encoders, decoders, and arithmetic circuits with minor errors; demonstrates functional hardware implementation with minimal guidance.	Demonstrates basic understanding of combinational logic; can implement simple circuits with significant guidance; makes frequent errors in circuit design or hardware interfacing.	Lacks fundamental understanding of combinational logic; unable to design or implement functional circuits; shows major misconceptions and inability to apply theoretical knowledge to practical tasks.

Sequential logic circuits: flip-flops, counters, and shift registers	1. Can explain the operation of flip-flops and sequential circuits. 2. Can analyze timing and state changes. 3. Can design and implement counters and shift registers.	Demonstrates comprehensive understanding of sequential logic and timing; independently designs, implements, and tests flip-flop-based circuits, counters, and shift registers; accurately analyzes circuit behavior.	Shows solid understanding of sequential logic; can design and implement functional counters and shift registers with minor errors; demonstrates good debugging and problem-solving skills.	Demonstrates basic understanding of flip-flops and sequential circuits; can implement simple counters or shift registers with frequent errors and significant guidance.	Lacks fundamental understanding of sequential logic; unable to implement functional flip-flop, counter, or shift-register circuits; shows major misconceptions.
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Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	Good
B	3,0	80-84	
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	
C-	1,67	60-64	Satisfactory
D+	1,33	55-59	
D	1,0	50-54	
FX	0	25-49	
F	0	0-24	Fail

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- The student has answered all questions from instructor(/lecturer), with all responses being accurate and comprehensive. - The work(assignment) was submitted on time in the <i>Moodle system</i>. - The work(assignment) is well-organized, with clear and precise presentation. - Student Can explain their work clearly and competently.
80-89	- The student completed the work correctly but did not upload it to the <i>Moodle system</i> on time. - The student was unable to answer one of the instructor's questions. - There are minor errors in the work (such as grammatical mistakes). - The format of the student's submission is not well-presented and difficult to read for others.
70-79	- The student was unable to complete all tasks specified in the lab or practical work. - The student made mistakes in the tasks. - The student provided an incorrect answer to a question from the instructor.
60-69	- The work was uploaded only at the end of the trimester. - The work contains incorrect answers. - The work is not written in English.

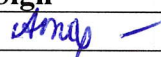
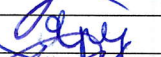
50-59	- The student was unable to answer more than three questions from the instructor during the defense - Half of the work is missing.
25-49	- The student completed only a small portion of the work. - The student refused to answer the instructor's questions.
0-24	- The student did not upload or submit the work. - The student was absent from classes without a valid reason. - The student was absent or did not complete the work but failed to inform the instructor before the final exams.

5. Course assessment methodology

Assessment is conducted continuously throughout the course, comprising 60% continuous assessment throughout the semester and 40% summative assessment during the examination session, totaling 100%. Continuous assessment evaluates students' ongoing performance through attendance, laboratory work defense, in-class activities, course assignments as well as independent study (SIS).

- **IASS (Instructor assisted student study)** - Instructors assist students in achieving success by delivering support, feedback, and resources. This support might take the form of clarifying difficult topics, recommending additional materials, or providing individualized guidance.
- **SIS** – Student engagement in self-directed study on assigned topics and the completion of relevant assignments given in the syllabus
- **Mid-term and End-term** are a review of the topics covered and an assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** for the course "Digital Logic Design" includes a final examination in the form of a computer-based exam or written paper-based exam or project.

6. Lecturer (lecturers) signature

Full name	Job title	Date	Sign
Alisher Amirov	Senior Lecturer	01/09/2026	
Alisher Batkuldin	Senior Lecturer	01/09/2026	
Aigerim Kalikova	Senior Lecturer	01/09/2026	
Didar Yarulin	Senior Lecturer	01/09/2026	
Olzhas Konakbayev	Senior Lecturer	01/09/2026	

